NATIONAL UNIVERSITY OF SINGAPORE
SCHOOL OF COMPUTING
MID-TERM TEST FOR
CS3244: MACHINE LEARNING

This is a sample set of answers for the midterm exam.

The email disclosing your answers may be for a different ordering of both the questions and answer options; as we needed to create 10 versions of the test to allow you all to take the exam next to each other in the LT 19 venue without worrying about copying.

Thus the correct answer options indicated on your email may differ from those indicated correct in this paper.

Multiple Choice Questions are scored in a binary (full marks / no marks) fashion. For Q23, there were multiple correct choices, as the answer fell on a border; so both have full marks.

Multiple Response Questions are fractionally scored as the number of correct values per binary letter choice, corrected for random chance guessing.

That is, we view each of the MRQs as 5 binary questions (e.g., Yes "A" or No "A"). But by random chance, we would expect to get 2.5 of these 5 questions correct. We thus need to correct for random guessing. We do this by subtracting 2 (rounding down from the appropriate value of 2.5) from your total of binary questions that you got right. Thus the maximum value would be 3 of 3 binary questions, for full marks for that question and either a 2/3rds or 1/3rd partial marks for other cases.

E.g., if the response key is "AB" and you answered "ABC", you would have gotten "A, B, no D, no E" correct and the binary question for C wrong (4 of 5 binary questions on each of the letters correct); 4 correct responses. After the random correction of subtracting 2 questions, we obtain 4-2 = 2 of the possible 3 binary questions correct.

In the assessment, each of the MRQs was worth 2 marks.

Here is a worked examples of the scoring (assuming a correct answer is "AB" worth 2 marks):

Sample Response	# of 5 binary questions correct	After –2 random guess correction	Marks for the question (given MRQ is worth 2 marks)
CDE	0	0 of 3 possible = 0	0
CD	1 (no E)	0 of 3 possible = 0	0
C	2 (no D, no E)	0 of 3 possible = 0	0
AC	3 (A, no D, no E)	1 of 3 possible = 1/3	0.66
ABC	4 (A, B, no D, no E)	2 of 3 possible = 2/3	1.33
AB	5 (A, B, no C, no D, no E)	3 of 3 possible = 3/3	2.0

Q1: Exam Identity (Part 1)

Please shade this MCQ as Ⓐ ● Ⓒ Ⓓ Ⓔ. This is an ungraded item. This is mandatory to ensure that your midterm is graded against the appropriate key.

Section 1. Multiple Choice and Multiple Response Questions.
Each question is worth 2 marks; 15 questions. Total: 30 marks.

Q2-5: Learning Paradigms

These responses pertain to the questions below:

A. Recognizing whether fast food is present in a picture
B. Transcribing the dialog between two specific people from the acoustic signal from microphones in a noisy MRT platform.
C. Planning module lecture times to minimize conflicts for students.
D. Answering multiple choice questions from midterms
E. Guessing which NUS courses a student takes in a current semester, based on their past transcript (modules and grades).

Mark all of the responses that could be construed as instances of the ML paradigm below. If none fit, do not mark any.

Q2. Supervised Classification. **ABDE**
Q3. Supervised Logistic Regression (Binary or Multinomial). **ABDE**
Q4. Unsupervised Machine Learning. *<none>*
Q5. Reinforcement Learning. **C**

Q6. The notion of a cost function is applicable to which tribe of machine learning? Mark one response.

A. Bayesians.
B. Connectionists.
C. Symbolists.
D. Evolutionaries.
E. Applicable to more than one tribe.

Q7. What is the minimum and maximum number of 2-D training points in 1 nearest neighbor that could be responsible for the following binary classification borders with 2 input dimensions? Assume that training examples have distinct input values. Mark one response.

A. 10, ∞
B. 9, ∞
C. 10, 10
D. 8, 10
E. 9, 9

Q8. Which of the following is **true** of *Noisy Targets*? Mark all responses that apply. If none fit, do not mark any.

A. We use the deterministic target function $f(x)$.
B. Noise = $y - f(x)$.
C. $P(y|x)$ is used instead of $f(x)$.
D. Noisy target = $\mathbb{E}[y|x] + (y - f(x))$.
E. None of the above.

Q9. Which of the following is **true** for *cost functions*? Mark all responses that apply. If none fit, do not mark any.

A. It measures the amount of error made by our hypothesis.
B. Which cost function to use depends on the problem.
C. Plausible measures are easy to use but cannot be proved.
D. Convenient measures have analytic proof.
E. Squared error is an example of a plausible measure.

Q10. Which of the following applies to PLA in the general case? Mark all responses that apply. If none fit, do not mark any.

A. A plausible stopping criterion is that we terminate after reaching a threshold cost value.
B. A plausible stopping criterion is that we terminate after a fixed number of iterations (large enough to converge).
C. In each iteration, we update the weight based only on misclassified example(s).
D. It will never reach a state where all examples classified correctly, if the dataset is not linearly separable.
E. After updating the weights, the accuracy of PLA will stay the same or improve.

Q11. How many marks a student can get in a competitive exam, based on her/his hours of study can be solved using a _____ model. Mark one response.

A. Linear Regression.
B. Logistic Regression.
C. Naïve Bayes.
D. None of the above.
E. A, B and C.

Q12. In logistic regression, we often use the negative of the log-likelihood of the training data as the cost function. During training, when we get a lower negative of the log-likelihood, it indicates which of the following? Mark one response.

A. The logistic regression model now fits the training data worse.
B. The logistic regression model now fits the training data better.
C. We will get a better classification accuracy in the training data.
D. We will get a better classification accuracy in the testing data.
E. The cost function should maximize the negative of the log-likelihood, not minimize it.

Q13. Suppose we are using the batch gradient descent (GD) to optimize the logistic regression model. We observe that it takes approximately T seconds to run each iteration of the GD. If we change the optimization method to stochastic gradient descent (SDG), how long it will take (approximately) to run one iteration then? Recall we use the variables m and n to denote the number of training instances, and the number of features per instance. Mark one response.

A. $\frac{T}{n}$

B. $\frac{T}{m}$

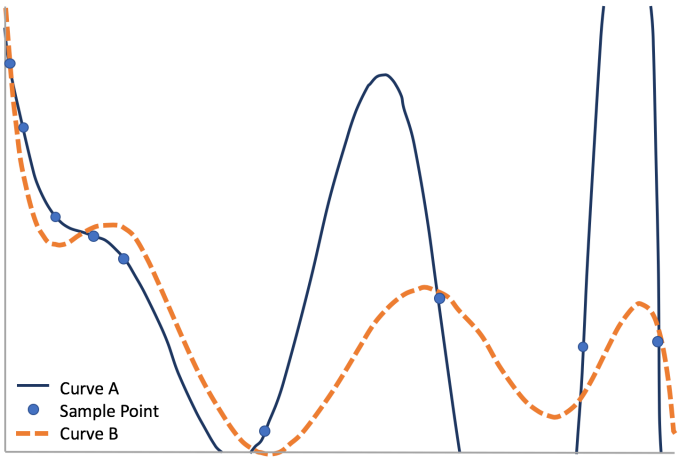
C. $\frac{T}{\log n}$

D. $\frac{T}{log\ m}$

E. None of the above.

Q14. From the following graph, identify which is target function and which is the approximation function. Note that the data points are noisy. Mark all responses that apply.

A. Target function: Curve A.
B. Approximation function: Curve B.
C. Target function: Curve B.
D. Approximation function: Curve A.
E. Impossible to tell.



Q15. To overcome *underfitting*, which of the following is/are helpful? Mark all responses that apply.

A. Having more data points.
B. Increasing the complexity of hypothesis.
C. Decreasing complexity of the target function.
D. Adding noise.
E. None of them.

Q16. If k is increased as in k-fold cross validation, the learned hypothesis h will tend more towards which of the following phenomenon? Mark one response.

A. Overfitting.
B. Underfitting.
C. Neither overfitting nor underfitting.

Q17: Exam Identity (Part 2)

Please shade this MCQ as Ⓐ Ⓑ ● Ⓓ Ⓔ. This is an ungraded item. This is mandatory to ensure that your midterm is graded against the appropriate key.

Section 2. Computation Questions. Marks differ per question. Total: 30 marks.

Q18-21: (16 marks) Linear Regression with Regularization
In this problem we are working on linear regression with regularization on points in a 2-D space. Figure 2 plots linear regression results on the basis of three data points, (0,1), (1,1) and (2,2), with different regularization penalties.

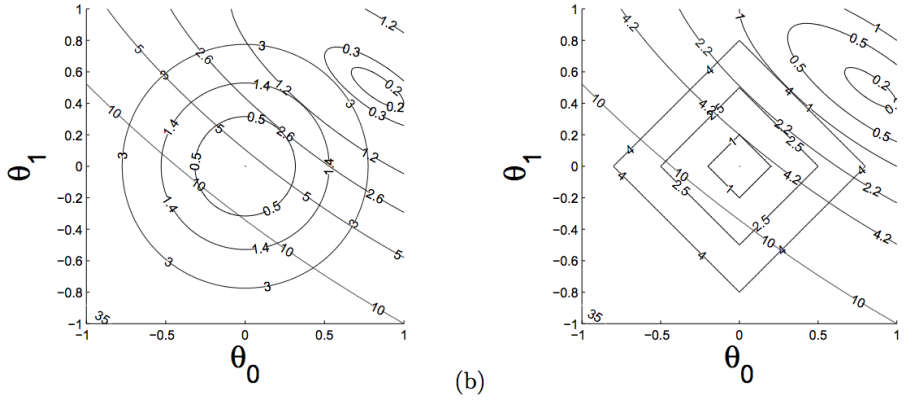


Figure 1. Contour plots for the linear regression problem with (a) L^2 or (b) L^1 regularization where the ellipsis correspond to the squared error term, and circles/diamond correspond to the regularization penalty term.

Assign each plot in Figure 2 to one (and only one) of the following regularization methods identified by Q18-21 (a one-to-one mapping). You may find Figure 1 helpful.

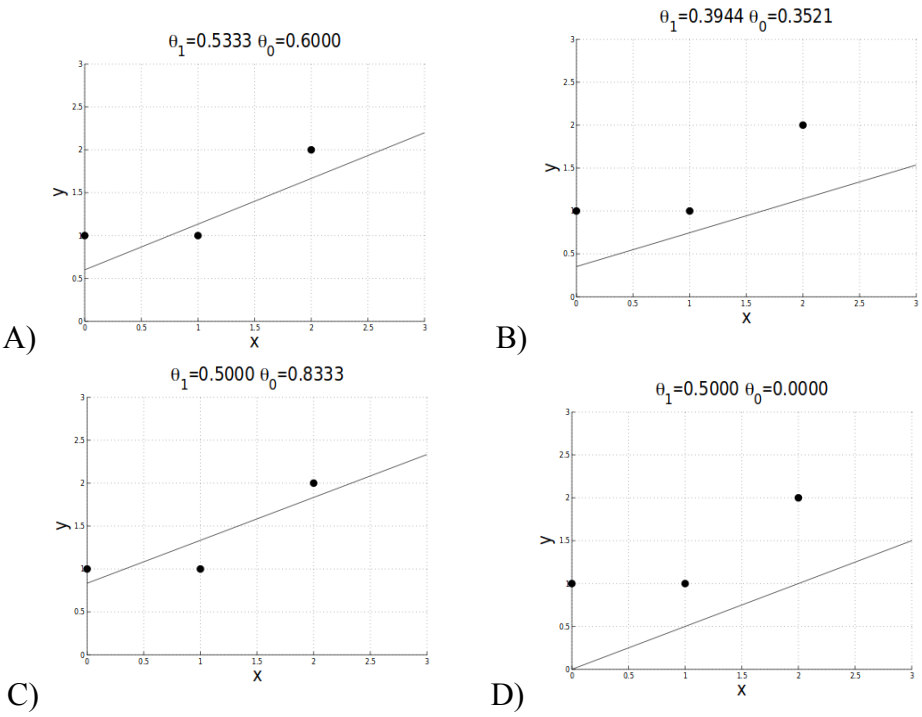


Figure 2. Linear regression with and with regularization.

Q18. (3 marks) No regularization. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2$ **C (.5, .83)**

Q19. (5 marks) L^2 reg. with λ = 5. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2 + \lambda(\theta_1^2 + \theta_0^2)$, where $\lambda = 5$. **B (.39,.35)**

Q20. (3 marks) L^1 reg. with λ = 5. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2 + \lambda(|\theta_1| + |\theta_0|)$, where $\lambda = 5$. **D (.5,0)**

Q21. (5 marks) L^2 reg. with λ =1. *I.e.,* $\sum_{j=1}^{3}\left(y_j - x_j\theta_1 - \theta_0\right)^2 + \lambda(\theta_1^2 + \theta_0^2)$, where $\lambda = 1$.
A (.53,.6)

Notice how the regularization makes the hyperplane more smooth (with a more horizontal slope and in the cast of L1, setting the intercept to 0.

Q22-Q23: Naïve Bayes (14 marks)

The following questions refer to this data table of instances with categorical inputs and output, which tabulates the temperature on the week before the midterm (binary outcome: Hot or Cold), whether a rare Pokémon was sighted earlier at the Arts canteen (binary: Yes or No) and whether the CS 3244 staff went to go to the Arts Canteen for their lunch (binary: Yes or No).

Temperature	Rare Pokémon Sighted at Arts?	Eat at Arts?
Hot	Yes	Yes
Cold	Yes	Yes
Cold	No	Yes
Hot	No	No
Cold	No	Yes

Recall that Naïve Bayes determines $P(y|X) = \prod_{j=1}^{n} P\left(x_j\middle|y\right)P(y)$. That is, it makes an independence assumption between the features of X, and calculates the likelihood as a term.

Q22: (6 marks) Today it's hot and rare Pokémon have not been sighted at Arts. Is it more likely that the CS3244 staff go to eat at Arts? Mark one choice.

P(temp=H|Eat_Arts=T)*P(Rare=F|Eat_Arts=T)*P(Eat_Arts=T) = 0.25 * 0.5 * 0.8 = 0.1
P(temp=H|Eat_Arts=F)*P(Rare=F|Eat_Arts=F)*P(Eat_Arts=F) = 1.0 * 1.0 * 0.2 = 0.2

A. Yes. More likely to go to Arts.
B. No. More likely to go somewhere else.
C. Unsure. There's an equal probability.
D. Impossible to know, given the data.

Q23: (8 marks) Today it's hot and some students saw the CS3244 staff go to the Arts Canteen for lunch. Calculate the probability that rare Pokémon have been sighted in Arts. Mark one choice.

#1 P(temp=H|Rare=T)*P(Eat_Arts=T|Rare=T)*P(Rare=T) = 0.5 * 1.0 * 0.4 = 0.200
#2 P(temp=H|Rare=F)*P(Eat_Arts=T|Rare=F)*P(Rare=F) = 0.33 * 0.66 * 0.6 = 0.133
#1/(#1+#2) = .2/(.13+.2) = 0.6 = 60%
Both answers of B and C are marked as correct.

A. Between 0.0 to 0.4.
B. Between 0.4 to 0.6.
C. Between 0.6 to 0.8.
D. Between 0.8 to 1.0.
E. Not possible as there's no causal relation. Where CS 3244 staff go does not cause Pokémon to show up.